Nanoelectromechanical Spin-Torque Memory Using Magnetic Tunnel Junction Nanoprobes for Energy-Efficient Nonvolatile Storage

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I. ABSTRACT

We demonstrate a nanoelectromechanical spin-memory architecture that replaces conventional magnetic-field-driven recording with electrically controlled spin-transfer torque (STT) and spin-orbit torque (SOT). A magnetic tunnel junction (MTJ) nanoprobe injects spin-polarized electrons into a perpendicular magnetic recording medium, producing localized magnetization reversal through spin-transfer torque. Readout is achieved through tunneling magnetoresistance (TMR) between the magnetic nanoprobe and the recording medium, enabling entirely electrical write and read operations without external magnetic fields. Nanoscale magnetic probes fabricated by helium-ion focused ion beam milling exhibit deterministic switching of CoFeB-based perpendicular media while significantly reducing the write energy compared with conventional inductive recording. By eliminating the need for magnetic-field generation, the proposed architecture provides a scalable pathway toward ultrahigh-density, energy-efficient, and three-dimensional nonvolatile memory technologies.

II. INTRODUCTION

Conventional magnetic recording relies on inductive write heads that generate localized magnetic fields to reverse the magnetization of recording media. As recording densities continue to increase, producing sufficiently large and highly localized magnetic fields for high-anisotropy media becomes increasingly difficult, resulting in higher power consumption and fundamental scaling limitations. Spin-transfer torque (STT) and spin-orbit torque (SOT) provide an attractive alternative by transferring angular momentum directly from spin-polarized electrons to the local magnetization, thereby eliminating the need for external magnetic fields.

In this work, we demonstrate a nanoelectromechanical recording concept in which a magnetic tunnel junction (MTJ) nanoprobe simultaneously functions as a localized writer and magnetic sensor. Spin-polarized tunneling electrons generated by the nanoprobe switch the perpendicular magnetic medium through STT, while the magnetic state is electrically detected through tunneling magnetoresistance (TMR). The probe-to-media separation is maintained at the atomic scale, allowing efficient spin transfer with highly localized current injection and ultralow switching energy.

III. RESULT AND DISCUSSION

The magnetic nanoprobe and recording medium were fabricated from CoFeB-based perpendicular magnetic multilayers. The probe was engineered to possess a higher perpendicular magnetic anisotropy than the recording medium by optimizing the CoFeB and MgO layer thicknesses, thereby providing a stable reference magnetization during both write and read operations.

A programmable point-contact system integrated with a scanning probe microscope (SPM) enabled simultaneous electrical characterization and nanometer-scale positioning. During operation, the probe was approached toward the recording medium using a piezoelectric actuator while maintaining a constant bias current. A stable tunneling junction was established when the measured current reached the threshold corresponding to a probe-to-media spacing below approximately 3 Å.

The MTJ nanoprobe was fabricated by helium-ion focused ion beam (He-FIB) nanomachining, as illustrated in Fig. 1(a). Thin-film magnetic multilayers deposited on commercial silicon probes were precisely patterned using helium and neon ion beams to define a nanoscale magnetic junction while

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minimizing ion-induced damage. The resulting structure provides highly localized spin-current injection with excellent dimensional control.

Figure 1(b) demonstrates localized magnetization reversal induced by spin-polarized tunneling currents. The programmed current sequence is in excellent agreement with the magnetic force microscopy (MFM) images acquired after each write operation, confirming deterministic STT switching of the perpendicular recording medium. Magnetization reversal occurs only when the applied current exceeds the critical switching threshold, providing direct experimental evidence that information can be written exclusively through spin-transfer torque without employing external magnetic fields. The strong spatial confinement of the tunneling current enables highly localized recording while substantially reducing the switching energy.

Figure 1. (a) Fabrication process of the MTJ nanoprobe using helium-ion focused ion beam milling. (b) Demonstration of localized STT writing. The upper panel shows the programmed current sequence, the middle panel illustrates the corresponding magnetization orientation, and the lower panel presents the MFM images confirming deterministic magnetization reversal. Scale bar: 50 nm.

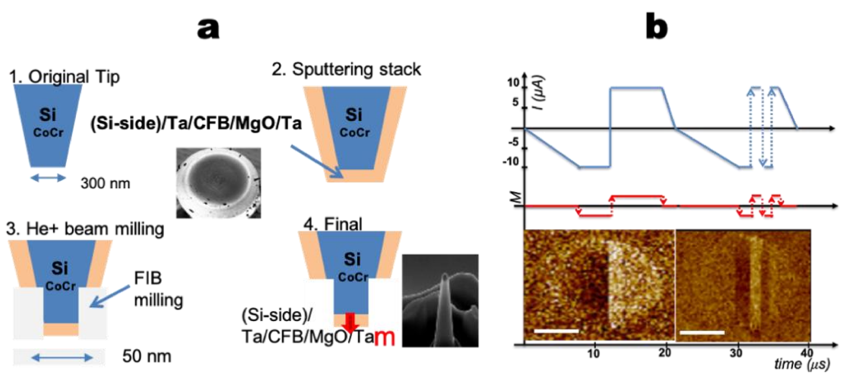

Figure 1. (a) Schematic of the nanoelectromechanical spin-memory system and fabrication of the MTJ nanoprobe writer using focused ion beam nanofabrication. (b) Experimental demonstration of localized writing through spin-transfer torque (STT). The programmed spin-polarized tunneling current produces deterministic magnetization reversal, as confirmed by magnetic force microscopy (MFM). Scale bar: 50 nm.

IV. CONCLUSION

We have demonstrated a nanoelectromechanical spin-memory architecture based on an MTJ nanoprobe that enables localized writing through spin-transfer torque and electrical readout through tunneling magnetoresistance. By replacing magnetic-field-driven recording with spin-polarized tunneling currents, the proposed approach significantly reduces write energy while overcoming the scaling limitations of conventional inductive recording. The combination of atomic-scale probe positioning, localized spin-current injection, and field-free operation offers a promising platform for ultrahigh-density magnetic recording, three-dimensional nonvolatile memories, and future spintronic information-storage technologies.

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